Binary Quantization Vision Transformer for Effective Segmentation of Red Tide in Multispectral Remote Sensing Imagery

Yefan Xie, Xuan Hou, Jinchang Ren, *Senior Member, IEEE*, Xinchao Zhang, Chengcheng Ma, and Jiangbin Zheng

***Abstract*— As a global marine disaster, red tides pose serious threats to marine ecology and the blue economy, making their monitoring crucial for preventing harmful algal blooms (HABs) and protecting the marine environment. In this study, satellite remote sensing was utilized to provide timely, large-scale, and continuous observation capabilities, overcoming the high cost and spatial and temporal limitations of in situ monitoring. However, existing remote sensing-based methods often exhibit coarse segmentation granularity and suffer from high computational complexity. To overcome these challenges, we propose a novel bimodal multispectral dynamic offset binary quantization visual transformer (DoBi-SWiP-ViT) that utilizes the ViT for global feature aggregation and parameter quantization for efficient segmentation. With the bimodal Swin-ViT with unified perceptual parsing (UPP) architecture, our model integrates data from multiple spectral bands to achieve fine-grained segmentation of large-scale remote sensing images. Additionally, we introduce a dynamic magnitude offset binary quantization ViT block to reduce the parameter redundancy and improve the computational efficiency. In addition, we validated the performance of our model through extensive comparative experiments on high-resolution imagery datasets of sea surface red tides collected from different satellite platforms. The results show that our proposed DoBi-SWiP-ViT has significantly improved the mean accuracy (mAcc) of the segmentation results. For the two test areas acquired from different satellite platforms, the improvements are 8.78% and 10.18%, respectively. This has demonstrated the superior performance of our model in detecting the red tides from high-resolution visible images, highlighting its effectiveness in capturing complex patterns and subtle features in multispectral imagery.**

***Index Terms*— Binary quantization, multi spectral imagery, red tide, remote sensing, segmentation, Vision Transformer (ViT).**

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I. Introduction

As a global marine disaster, red tides pose significant risks to the marine ecology, aquaculture, and blue economy. Therefore, monitoring red tides is crucial for preventing and reducing the hazards of harmful algal blooms (HABs), which is essential for protecting the marine environment. Traditional on-site monitoring collects data on marine environmental elements through fixed-point observation [1] and mobile observation [2]. Although these methods tend to have local spatial and temporal continuity and high accuracy, they are limited by the reliability of sensors and the trustworthiness of data [3], [4]. Considering the diverse spatial distribution of red tides and their rapid rate of change, they often fail to meet the requirements of large-scale timely monitoring. Satellite remote sensing has the advantages of timely, large-scale, and continuous observation [5], [6], which is conducive to the rapid location of hazardous areas and impact levels of red tide, where the accurate location of such areas can also guide the ground staff to advance the response speed of protection and specific actions to mitigate the hazard [7].

Conventional remote sensing methods indicate the presence of red tide by identifying changes in water color caused by algal blooms, using ocean color data captured from platforms such as Landsat, MODIS, and Sentinel satellites. Indices such as the red tide index (RI) [7], P. donghaiense index (PDI), and diatom index (DI) [3], as well as a series of improved RI algorithms [9], [10], have been developed for this purpose. Alternatively, spectral analysis methods use specific spectral bands to detect chlorophyll-A, bio-optical properties of seawater, or fluorescence line height (FLH) as alternative indicators to determine the presence of red tides [11], [12], [13]. In recent years, deep learning methods have significantly advanced the intelligent interpretation of remote sensing images by fully leveraging their spectra, textures, and fine features [14], [15], [16], [17], [18]. The encoder–decoder architecture proposed in U-Net [19] has been beneficial, effectively capturing local features at different scales, enhancing spatial detail and structural

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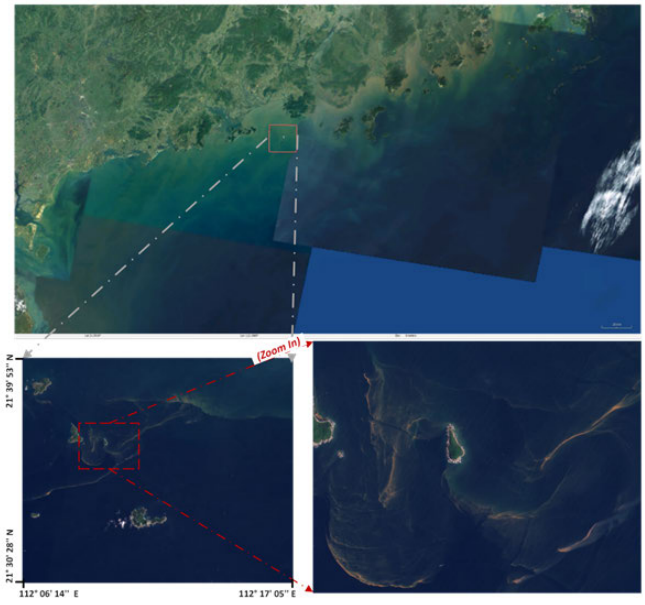


Fig. 1. Example of color imagery of the red tide in the monitoring area (area 2) from the PlanetScope satellite [8]. The top portion of the image is color imagery at a large scale, processed by stitching sequentially collected imagery by coordinate correction. The bottom left image is one of the imagery from the experimental dataset, with the red dashed box showing an example image after the cropped process.

recovery in images. However, the local-focused features limit its ability to represent the global context, which is crucial when processing large-scale images. Furthermore, the high parameter redundancy inherent in deep neural networks (DNNs) has led to to significant computational costs [14], [20].

To address these challenges, we propose a binary quantization Vision Transformer (ViT) for red tide segmentation in multispectral satellite imagery, by using a feature fusion scheme with a unified perceptual parsing (UPP) architecture to further enhance the fine feature extraction capability of the model. To address the issues associated with the large model complexity of the ViT structure and the unified perceptual structure, we propose a dynamic magnitude offsets binary quantized ViT (DoBi-ViT) block structure to reduce the parameters. Moreover, we conducted extensive comparative experiments on high-resolution imagery datasets of sea surface red tides collected from various satellite platforms (as shown in Fig. 1) to validate our method. The results demonstrate the superior performance of our model in red tide segmentation, highlighting its ability to capture complex patterns and subtle features in multispectral imagery.

The major contributions of the model can be highlighted as follows.

1) We propose a bimodal ViT with UPP architecture, significantly enhancing the ViT to extract fine-grained semantic details and improve its performance in high-resolution and large-scale scenes.
2) To address the model complexity associated with ViT structures and unified perceptual architectures, we have designed a dynamic offset binary-ViT (DoBi-ViT) block structure. This design reduces the overall parameter footprint of the model and enhances its efficiency.
3) A high-resolution dataset of red tide imagery was collected from three public satellite platforms. Unlike existing methods that crop and split data from single imagery, it includes scenes from different sea areas, times, and outbreak scales for experimental training and validation. Our model demonstrated superior performance and was validated through extensive comparative experiments.

II. Related Work

A. Deep Learning-Based Red Tide Segmentation

Red tide segmentation and monitoring methods can be broadly categorized into in situ surveys and remote sensing-based techniques [21]. Given its extensive spatial coverage and short revisit intervals, remote sensing technology has become a pivotal tool for red tide monitoring and segmentation [22], [23], [24]. Due to the sensitivity of spectral-based monitoring methods to the associated segmentation thresholds and the advancements in ground resolution of remote sensing, various deep learning techniques have been applied for red tide segmentation [25]. Jiang et al. [20] employed a deep confidence network model to detect red tide using airborne hyperspectral remote sensing data. Li et al. [15] proposed a red tide extraction method based on deep learning with autonomous aerial vehicle (AAV) remote sensing images. Lee et al. [26] combined the high-loss sample mining method with the ResNet and geostationary ocean color imager (GOCI) image data for red tide segmentation. Zhao et al. [25] proposed a red tide segmentation method based on the U-Net using HY-1D satellite coastal zone imager (CZI) data. Shen et al. [27] proposed a progressive CNN-transformer alternating reconstruction network (PCTARN), which introduces the global–local dynamic priors and stacks lightweight convolutional modules at different levels to efficiently enhance the reconstruction quality of red tide hyperspectral data, thereby facilitating red tide species identification. However, existing methods often encounter challenges such as insufficient emphasis on the global features, excessive granularity in semantic segmentation, and parameter redundancy, which hinder both the efficiency and accuracy of red tide monitoring. To address these issues, we propose an alternative approach that reduces the number of parameters in ViT-based models through parameter quantization. This approach significantly reduces model parameters while maintaining the strong global feature extraction capability, enabling high-precision red tide segmentation in large-scale remote sensing images.

B. Vision Transformer

The Transformer was initially proposed for machine translation tasks [28]. In the field of natural language processing (NLP), Transformer-based approaches have achieved state-of-the-art (SOTA) performance across various tasks. Around the same period, before the introduction of Transformer architecture into the field of computer vision (CV), researchers had already recognized the potential of attention mechanisms for enhancing the capabilities of neural networks. These

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efforts included employing self-attention to improve the performance of conventional convolution neural networks (CNNs) by enabling them to adaptively focus on features of interest within an image. In [29], skip connections with additive attention gates were integrated into a U-shaped architecture for medical image segmentation. However, the core component of the model was still based on convolutional constructions, indicating that it remained fundamentally CNN-based.

Driven by the success of the Transformer across various fields and the application of attention mechanisms in CV, a pioneering ViT was introduced in [30], marking the first image recognition model built upon the Transformer mechanism. Compared with CNN-based methods, a notable drawback of ViT is that it requires pre-training on its own large dataset. To mitigate the challenges associated with training ViT, Deit [31] proposed several training strategies that enable ViT to perform effectively on ImageNet. Recently, several additional works have been made based on ViT [32], [33], [34]. Among these, it is worth highlighting that an efficient and effective hierarchical ViT, called Swin Transformer, is proposed as a vision backbone in [32]. Leveraging the shifted windows mechanism, the Swin Transformer achieved SOTA performance in various vision tasks, including image classification, object detection, and semantic segmentation. Swin-UNet [35] is the first pure Transformer-based U-shaped architecture, comprising an encoder, bottleneck, decoder, and skip connections. The Swin Transformer block constitutes the core of the encoder, bottleneck, and decoder. The process begins by dividing input images into nonoverlapping patches, each treated as a token. These tokens are processed by the Transformer-based encoder to extract deep feature representations. The decoder, equipped with a patch-expanding layer, upsamples the extracted features and integrates them with multiscale features from the encoder via skip connections. This integration restores the spatial resolution of the feature maps, enabling precise segmentation. However, a common limitation of ViTs is their high computational demand, which can pose a bottleneck in resource-constrained environments.

C. Parameter Quantization

With the advancement of DNNs, the number of parameters and computational costs has grown substantially. To tackle the challenges of deploying large models on resource-constrained platforms, parameter quantization has emerged as a widely adopted solution. This technique compresses DNNs by replacing weights and activations with low-bit representations, enabling significant reductions in model size while preserving the original network structure [36], [37], [38], [39], [40], [41], [42], [43].

The binary quantization paradigm represents an extreme form of parameter quantization, where both the weights and activations in DNNs are constrained to 1-bit representations. Compared to their full-precision counterparts, binary quantization replaces multiplication operations with bitwise operations, offering the potential to reduce network size by a factor of 32. Xnor-Net [42] proposed that a real-valued scaling factor could be implemented to each output channel of the binary convolution for accuracy improvement, which has become a common practice for binary networks. Bi-real-Net [44] argued that the real-valued skip connection presents the basis of binary networks, and they suggested converting the downsampling layer to full precision values, trading negligible computational complexity for improved accuracy. Xnor-Net++ [45] proposed using PReLU to smooth the gradient approximation. Wang et al. [46] proposed leveraging reinforcement learning to model channel correlations, enabling better preservation of the sign output of the convolution. Ding et al. [47] introduced a set of regularizers into the loss function to constrain activation values and ensure proper gradient flow. Alizadeh et al. [48] perform validation tests on the impact aspects of gradient clipping and batch-norm momentum. Xu et al. [49] proposed using a rectified clamp unit (ReCU) to leverage the information entropy and quantization error relationships in the binary neural network (BNN). RB-Net [50] was proposed to achieve a balance between the accuracy and efficiency in object classification tasks by introducing reshaped point-wise convolution (RPC) and integrated balanced activation (BA). Despite these advancements, binary quantization continues to face significant accuracy degradation, particularly in pixel-level segmentation tasks. This will be addressed in our model as detailed in Section III-B.

III. Proposed Method

The overall structure of the proposed bimodal multispectral dynamic offset binary quantization ViT (DoBi-SWiP-ViT) segmentation model is illustrated in Fig. 2. At the input stage, similar to the conventional visual DNNs, a cropping operation is first applied to the collected imagery. The cropped image serves as the model input, and following feature aggregation through the binary-quantized ViT backbone with dynamic offsets, the features are fused and processed at multiple levels via the UPP module. This process culminates in the generation of the final semantic segmentation results. The detailed implementation of the key modules is described below.

A. Bimodal Swin-ViT With UPP

The decision to use a Transformer as the primary feature extractor is motivated by its robust capability to capture global features, making it particularly well-suited to tasks such as red tide segmentation, where accurately identifying key features across the entire image is crucial. Conventional ViTs utilize multihead self-attention (MSA) mechanisms for global feature aggregation. However, the computational complexity of MSA increases quadratically with the length of the input sequence ($O(n^2 \cdot d)$, where n is the sequence length and d is the feature dimension. This imposes a substantial computational burden, particularly when processing large-scale datasets or high-resolution imagery. Therefore, the adoption of the Swin Transformer is driven by its ability to achieve effective feature aggregation while markedly enhancing computational efficiency. The Swin Transformer introduces a window-based MSA (W-MSA) mechanism, which confines attention computations to nonoverlapping local windows. This approach significantly reduces computational complexity, particularly when the window size M is considerably smaller than the input

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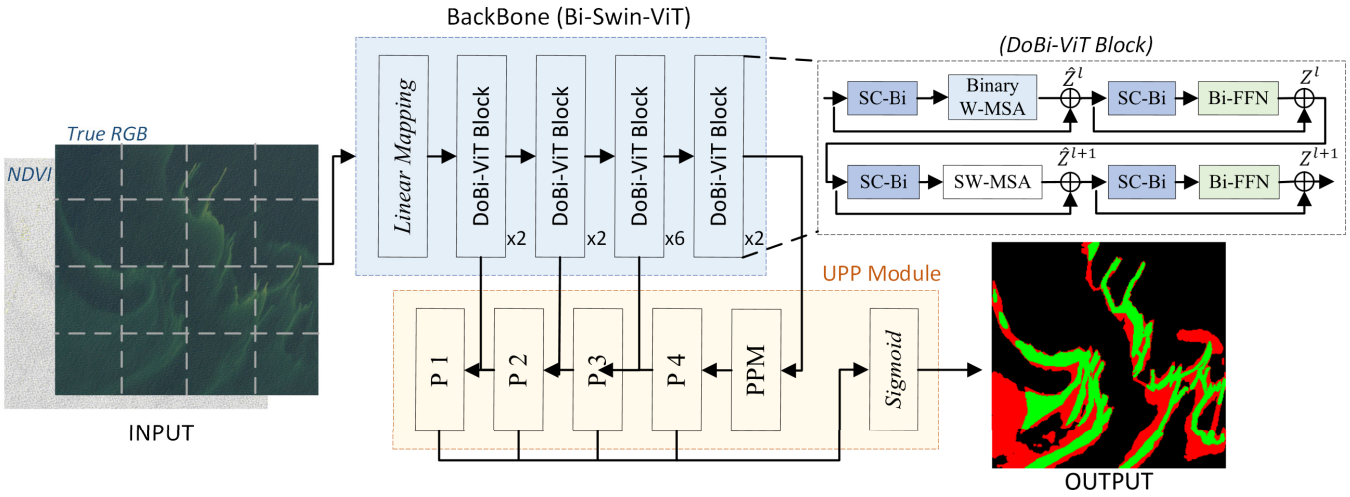


Fig. 2. Overall structure of the DoBi-SWiP-ViT segmentation model. In the UPP module, the PPM has four different pyramid scales with layer sizes of 1×1, 2×2, 3×3, and 6×6, preserving global context information at multiple scales. P1, P2, P3, and P4 denote multilevel feature maps of (1/4, 1/8, 1/16, 1/32) scales, which are derived through upsampling, downsampling, and convolution operations on the feature maps extracted from corresponding layers of the ViT. The processed feature maps are then combined element-wise to produce the fused feature map for the subsequent layer. Additionally, the Bi-FFN in the DoBi-ViT block comprises two binary linear projection layers and a single activation layer.

dimension n, yielding a reduced complexity of $O(n^2 \cdot d/M^2)$. This represents a substantial improvement over conventional MSA. The reduction in complexity not only improves the computational efficiency of the model but also enables it to handle large-scale visual tasks more effectively by accommodating larger input sizes.

The remote sensing image acquisition sensor on satellite platforms is equipped to capture multispectral data, including visible and near-infrared (NIR) bands. In particular, the NIR bands offer critical insights into the physical and biological properties of water bodies that are imperceptible through standard visible spectroscopy. Specifically, NIR wavelengths have the capability to penetrate water to a certain depth, enabling the detection of subtle variations in water composition and temperature that serve as indicators of red tide events [51]. Furthermore, the NIR band exhibits high sensitivity to chlorophyll and other pigments associated with algal blooms, which are the primary components of red tides [52]. Building on this, the model design proposed in this article incorporates a bimodal multispectral input structure within the traditional Swin-ViT architecture. This design leverages bimodal data sources derived from different spectral bands of multispectral images, utilizing satellite-acquired multispectral information to enrich the feature representation of visible images for red tide monitoring. By adopting this dual-input approach, the model effectively captures additional physical characteristics of the scene, thereby enhancing its overall feature representation capability.

Building upon this foundation, the design of the multispectral data input facilitates enhanced feature extraction by leveraging dual input data, enabling the model to integrate features from multiple spectral bands for a more comprehensive understanding of the scene. The proposed dual-stream Bi-Swin-ViT architecture processes these multispectral inputs in a coordinated manner, improving the model's robustness to variations in input data. In addition to combining the red, green, and blue (RGB) bands to synthesize true RGB images, the normalized difference vegetation index (NDVI), specifically designed for detecting HABs in water bodies, is incorporated to provide supplementary feature information. The calculation of NVDI involves the irradiance of the red and NIR bands, as shown in (1). Under the multiinput scenario, data augmentation methods applied to images from different spectral bands are maintained consistently to ensure uniformity across modalities

$$\mathrm{NDVI} = (B_{\mathrm{NIR}} - B_{\mathrm{Red}})/(B_{\mathrm{NIR}} + B_{\mathrm{Red}}). \tag{1}$$

In the backbone component utilizing the ViT, the input image is treated as a sequence of tokens by dividing it into small blocks and embedding positional information during preprocessing. At this stage, the input image is partitioned into multiple fixed-size patches, which are linearly projected to form a sequence before being fed into the Transformer. In this article, the patch size is set to 4, and the window size is set to 7. This approach enables each patch to capture information from the entire image, rather than being constrained to its local region. Additionally, the positional encoding process enhances the model's capacity to perceive global information. The above processing steps for the input image are formulated as in the following equation:

$$\mathbf{z}_0 = \left[\mathbf{x}_{\mathrm{class}}; \mathbf{x}_p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}; \cdots; \mathbf{x}_p^N\mathbf{E}\right] + \mathbf{E}_{\mathrm{pos}} \tag{2}$$

where $\mathbf{z}_0$ denotes the initial input sequence. $\mathbf{x}_{\mathrm{class}}$ represents the class token, which encodes the class-specific information of the input sequence. $\mathbf{x}_p^i$ corresponds to the input feature at patch i. The feature embedding matrix $\mathbf{E}$ maps the input features to a high-dimensional space, with $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$. The positional embedding matrix $\mathbf{E}_{\mathrm{pos}}$, which represents positional information for each input position, is defined as $\mathbf{E}_{\mathrm{pos}} \in \mathbb{R}^{(N+1) \times D}$.

However, this mechanism results in the model lacking a sufficiently clear perceptual field for different patch blocks

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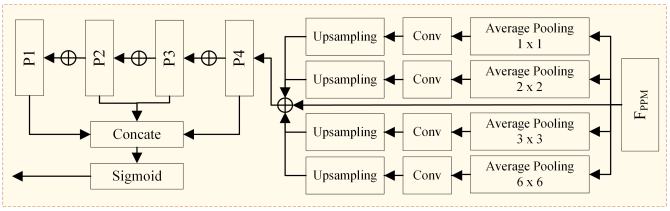


Fig. 3. Detailed structure of the UPP module employs a multiscale feature fusion mechanism, facilitating the flow of information between deep and shallow features through cross-level fusion operations.

during feature extraction, making it less effective than CNNs at understanding local feature information. To address this limitation, we introduce the UPP module in the feature extraction phase to mitigate the issue of local feature blurring as shown in Fig. 3. This is achieved through cross-level multistage feature fusion operations, ensuring effective information flow between deep and shallow features. Functionally similar to the traditional feature pyramid structure, its purpose is to extract high-level semantic features from input images via a multiscale feature fusion mechanism, enabling the capture of information representations across different scales.

Specifically, this architecture enhances the detail recovery capability and ensures the semantic representation consistency of multiscale features by incorporating a feature pyramid network (FPN) and a pyramid pooling module (PPM) following the feature extractor. The FPN enriches the semantic information of features through a top-down pathway and lateral connections, enabling the network to effectively handle objects of varying scales. Meanwhile, the PPM captures global context by pooling features across various regions, improving the model's comprehension of the background and large objects. Additionally, the integration of a global pooling operation within this module provides a global feature representation of the entire image. This fusion of global contextual information with local features provides richer semantic information and more accurate segmentation predictions. The concatenation and channel upsampling steps involve upsampling all pooled results to the same spatial dimensions, concatenating them along the channel-wise, and adjusting the channel dimensions via convolutional layers to align with the input requirements of subsequent layers.

In summary, we propose the bimodal Swin-ViT framework with a UPP module, designed to incorporate more diverse and effective information by integrating bimodal multispectral inputs and advanced feature parsing. This architecture maintains computational efficiency and maximizes the extraction and utilization of contextual information. By leveraging the Transformer mechanism and the UPP approach, the model effectively captures both global and local features, significantly enhancing semantic understanding through their seamless integration. This synergy enables the bimodal Swin-ViT and UPP module framework to deliver superior performance in image segmentation tasks, particularly in scenarios demanding high precision and adaptability.

B. DoBi-ViT Block

ViT models are characterized by hierarchical and partitioned self-attention mechanisms, which effectively capture

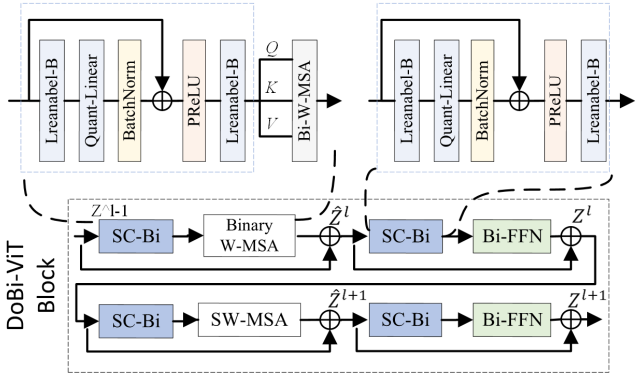


Fig. 4. SC-Bi design in a learnable DoBi-ViT.

global dependencies and contextual information across images. However, these models are computationally intensive, particularly in large-scale applications requiring timely processing. To address these challenges, we propose applying binary quantization to specific components within the ViT architecture. This approach aims to reduce computational complexity and memory usage while maintaining the model's ability to accurately represent complex image features. Implementing binary quantization accelerates inference speed and facilitates deployment on resource-constrained platforms.

BNNs primarily implement binary processing of the network structure for classification tasks [39], [44], [53], [54], which proves that the feature extraction backbone part of the network already has sufficient representational capacity. However, in semantic segmentation tasks, due to the strict requirement of the sensitivity of the parameter in the segmentation task, the vanilla binary segmentation network can lead to severe performance deterioration situation [55]. Furthermore, addressing the binarization of ViT models involves unique challenges due to their reliance on the attention mechanism to capture global information and their substantial parameter requirements during training. While the attention mechanism provides ViT models with strong representational capabilities for image tasks, simple binarization often compromises the accurate representation of complex attention weights. This results in a loss or blurring of global information, ultimately hindering the model's ability to effectively comprehend the overall image structure.

To address this challenge, we propose a learnable DoBi-ViT block structure, as illustrated in Fig. 4. This design introduces an additional SC-Bi module both before and after the W-MSA, enabling the binarized transformer blocks to achieve lower quantization error through the introduction of dynamic offset parameters. Compared to conventional binarization methods, this approach enhances suitability for high-precision segmentation tasks. Subsequently, the feature stream, processed via residual skip connections, serves as input to the binary feed-forward network (Bi-FFN), where further refined feature representations are extracted through the MSA mechanism.

For each matrix multiplication in the forward phase, a sign function is applied to each input activation X and weight W. A threshold vector σ_X is applied to the real-valued inputs prior

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to applying the sign function, allowing these inputs to account for distributional shifts. For the weights, the threshold $\mu(W)$ is determined by computing the mean value of all elements within the matrix, as suggested in [56], [57], and [58]. For the activations, the threshold parameter is optimized through backpropagation to minimize the task loss as in [59] and [60]. The matrix multiplication output $Y(X)$ in a binary ViT block is calculated as shown in the following: equation:

$$Y(X) = \frac{1}{n}||W||_1 \text{Rsign}(X) \otimes \text{sign}(W - \mu(W)) \tag{3}$$

where $\text{Rsign} = \text{sign}(X + \sigma_X)$ as described in [59]. $\otimes$ denotes the binary convolution, which can be implemented using bit-wise operations such as XNOR and Pop-Count.

In each binary fully connected layer (BiFC) of a binary transformer, we introduce a residual connection that directly links the input to the output of the linear layer, as shown in (4). This residual connection is designed to preserve information from the previous layer, consistent with the methodology of [44] and [59]. Furthermore, all layer normalization in the ViT model is replaced with batch normalization (BN) [61], since all linear layers have a normalization layer after it, as in [62]. This substitution facilitates faster inference and training compared to layer normalization

$$\text{BiFC(X)} = \text{RPReLU}(\text{BN}(Y(X)) + R(X)) \tag{4}$$

where X denotes the input of the layer, $R(\cdot)$ represents the residual connection, and $\text{BN}(Y(X))$ refers to the output of the linear layer. The $\text{RPReLU}(\cdot)$ activation function, as proposed by [59], is applied following each residual connection.

During the backpropagation process, we follow the principle of binary quantization and use the straight-through estimator (STE) [44], to approximate the derivative of the sign function with respect to the input, as presented in the following equation:

$$\frac{\partial \text{sign}(x)}{\partial x} = \begin{cases} 1 & \text{if } |x| \leq 1 \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Based on the aforementioned settings, we propose a learnable DoBi-ViT block structure within the SC-LB-Bi architecture. To address the issue of feature information collapse caused by the linear layer in the quantization model, we introduce a trainable bias (Learnable-B), which performs a sensitivity shift operation on the features during training. This adjustment enhances the diversity of quantization thresholds across different channels by transforming the single sign function into a soft-threshold sign function. Additionally, we adopt the rectified parameter exponential linear unit (RPeLU) activation function, which introduces further diversity to the hard-threshold quantization originally achieved by the sign function. These modifications improve the model's ability to handle threshold uniformity during training, enhancing its performance after quantization.

In the multi-head attention (MHA) mechanism, for each head h, the received input features are transformed into three branches: query (Q), key (K), and value (V), which are used for subsequent processing. In the Bi-W-MSA module of the binary transformer with N_H attention heads, the output from the BN is subsequently used to compute the query, key, and value matrices, denoted as Q_h, K_h, and V_h, where h represents as each attention head. The specific formulation is provided in (6). In this case, $Q_h, K_h, V_h \in \mathbb{R}^{(N+1)\times D_h}$, where D_h represents the dimensionality of the vectors in each head. Additionally, $D_h = D/N_H$, where D is the total dimensionality of the input feature representation and N_H is the number of attention heads

$$\begin{aligned} Q_h &= \text{BiFC}_{Q_h}(\hat{H}_h) \\ K_h &= \text{BiFC}_{K_h}(\hat{H}_h) \\ V_h &= \text{BiFC}_{V_h}(\hat{H}_h). \end{aligned} \tag{6}$$

The outputs from all heads are concatenated and processed by the fully connected layer, BiFC_{out}, to compute the MHA output [53]. These are additionally incorporated into the head output to retain the information from the query, key, and value matrices. A primary residual connection is applied to the output of the MHA, as described in the following equation:

$$\begin{aligned} F &= \text{BiFC}_{\text{out}}(\text{Cat}(B_1, \ldots, B_n)) + H \\ B_n &= \text{RPReLU}(\text{BN}(P_h \cdot \text{Rsign}(V_h)) + Q_h + K_h + V_h \\ P_h &= \alpha \cdot \lfloor \Theta\left(\text{Softmax}\left(\frac{\text{Rsign}(Q_h) \cdot \text{Rsign}\left(K_h^T\right)}{\sqrt{D_h}}\right), 0, 1\right)/\alpha \rceil \end{aligned} \tag{7}$$

where P_h denotes the attention matrix derived through the scaled dot-product operation. B_n represents the output of individual heads within the MHA. The learnable scaling factor α, optimized using the method outlined in [63], dynamically adjusts the output range and sensitivity. Specifically, α ensures that the attention weights remain balanced across varying data distributions, enabling the model to better adapt to the dynamic range of quantized data characteristics. The threshold function $\Theta(x, \rho_1, \rho_2)$ constrains the output to the interval defined by ρ_1 and ρ_2. To enhance the robustness and suppress the noise, the output undergoes discretization via the round-to-nearest-integer function $\lfloor \cdot \rceil$. This rounding mechanism minimizes the effect of minor numerical fluctuations on the attention weights, thereby improving the stability of the model during training and inference. The rounding mechanism further facilitates the model's focus on specific attention patterns, minimizing the impact of minor numerical fluctuations on the attention weights and enhancing robustness. This allows the model to balance different attention weights more effectively, thereby boosting overall performance.

Following the aforementioned operations, the residual output F is normalized through a BN layer. It is subsequently passed through a Bi-FFN layer, comprising two BiFC layers. Finally, a residual connection is applied to the Bi-FFN output, yielding $R = \text{Bi-FFN}(\text{BN}(F)) + F$, which serves as the input for the subsequent DoBi-ViT block.

IV. Experiments

A. Datasets

The datasets for this study were collected from the open-access Landsat-8 [64], Sentinel-2 [65], and PlanetScope [8]

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TABLE I
OVERVIEW OF SATELLITE SYSTEMS: SENTINEL-2, LANDSAT-8, AND PlanetScope

Satellite System	Sensor Name	Spectral Bands	Resolution	Orbital Altitude	Number of Satellites	Revisit Period
Landsat-8 [64]	Operational Land Imager (OLI) Thermal Infrared Sensor (TIRS)	OLI: 9 Bands TIRS: 2 Bands	OLI: 30m TIRS: 100m	705 km	1	16 days
Sentinel-2 [65]	Multi-spectral Imagery (MSI)	13 Bands	10m, 20m, 60m	786 km	2	5 days
PlanetScope [8]	Dove Satellites	8 Bands	3-5 m	400 km	Over 120	1 day

TABLE II
OVERVIEW OF SPECIFIC ACQUISITION AREA COORDINATE INFORMATION

Satellite	Resolution	Date	Imagery	Location	Usage
Landsat-8 [64]	30m/px	2020.08.18	LC08_L1TP_116037_20200818_20200920_02_T1	Yellow Sea	Train & Valid
Sentinel-2 [65]	10m/px	2020.08.15	S2A_MSIL2A_20200815T021611_N9999_R003	Yellow Sea	Train & Valid
		2021.02.14	2021-02-14-00_00_2021-02-14-23_59_Sentinel-2_L2A	Vietnam	Train & Valid
		2021.02.23	2021-02-23-00_00_2021-02-23-23_59_Sentinel-2_L2A	Vietnam	Train & Valid
		2020.08.15	Sentinel-2 L2A 2020-08-15-02	Yellow Sea	Test (Area A)
PlanetScope [8]	3m/px	2021.02.22	20210222_030742_62_227a	Guangdong	Train & Valid
		2022.04.10	20220410_021109_59_241f	Guangdong	Test (Area B)

satellite platforms. The sensor specifications and revisit cycles of these satellites are summarized in Table I. Among these satellites, Landsat-8 continues the decades-long tradition of the Landsat program, capturing high-quality and detailed surface features of the Earth. Equipped with advanced sensors, it provides critical data that support long-term environmental change studies. Sentinel-2, managed by the European Space Agency (ESA), consists of two satellites with multispectral imaging capabilities. These satellites deliver high-resolution imagery valuable for applications, such as vegetation monitoring, soil and water analysis, urban planning, and disaster management. Notably, Sentinel-2 is the first optical satellite series to incorporate three "red-edge" bands, offering crucial insights into vegetation health and conditions. Meanwhile, PlanetScope, operated by Planet Labs, consists of a constellation of over 120 "Dove" satellites. This system can image the entire land surface of the Earth daily, with a total acquisition capacity of 200 million square kilometers. It is particularly well-suited for rapid responses to natural disasters, agricultural monitoring, and urban development initiatives.

Considering the distribution characteristics of different objects in the study area, multiple locations at different times were selected to construct the training sample dataset, as summarized in Table II. The data were collected from diverse regions across different temporal periods, with the ground truth (GT) determined through visual interpretation. To prepare the dataset for model training, the images and their corresponding labels were divided into cropped images of size 512 × 512 pixels using a sliding window approach, matching the input size of the network. This process yielded a total of 143 samples, which were partitioned into training and validation datasets based on different imagery areas. To ensure sufficient training data and mitigate the risk of overfitting, data augmentation techniques, including horizontal, vertical, and diagonal flipping, were applied to the input images during the training phase. For testing, two separate imagery regions not included in the training dataset were selected. To maintain consistency with the training process, the same sliding window strategy was applied to the test images, dividing each image into cropped images of 512 × 512 pixels.

Additionally, the two test areas were selected from different satellite platforms to represent distinct red tide conditions, allowing for a comprehensive evaluation of the model's robustness across various scenarios and data sources. Specifically, Test Area A, originated from the Sentinel-2 satellite [65], serves as a critical baseline, utilizing independent remote sensing imagery that is temporally and spatially distinct from the training dataset, to assess the model's capability for generalized red tide detection. This approach contrasts with prior studies [66] that often relied on cropping the training and testing data from the same scene. Test Area B, sourced from the PlanetScope satellite [8] and characterized by distinct temporal and spatial conditions, was introduced to further examine the model's robustness and adaptability across diverse contexts. Experimental results indicate that our model consistently outperforms several SOTA methods across both test areas, demonstrating its superior robustness and generalization capability.

B. Metrics

The performance of the proposed method was evaluated using three criteria, including the mean intersection over union (mIoU), the mean dice coefficient (mDice), and pixel-based mean accuracy (mAcc). The mIoU evaluates the overlap between the model's predictions and GT labels by computing the ratio of intersection to the union of the predicted and GT regions. Meanwhile, mDice provides another measure of segmentation accuracy, reflecting the degree of overlap between predicted and GT regions. It computes the ratio of intersection to the average size of both regions. On the other hand, mAcc assesses the model's pixel-level classification accuracy by determining the ratio of correctly classified pixels to the total number of pixels. The calculation methods for these metrics are presented in the following equation:

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i}$$

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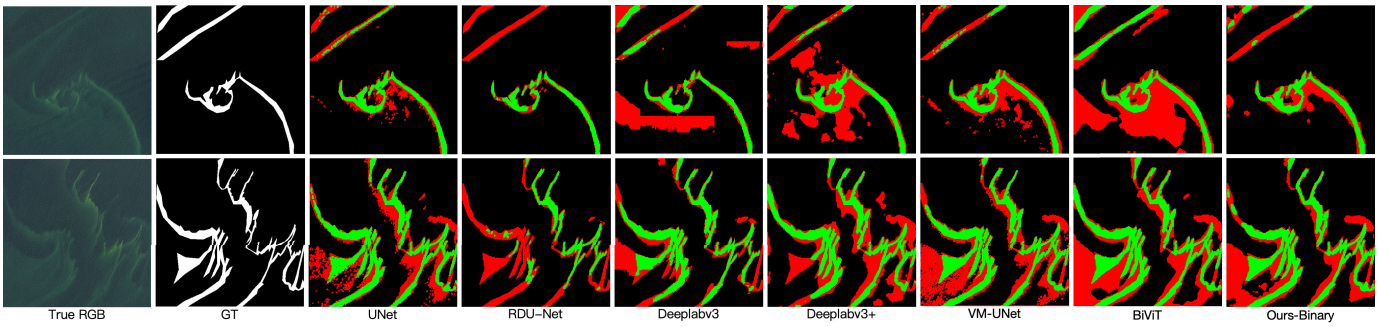


Fig. 5. Visualization of the comparison results from various methods on the test imagery (denoted as Test Area A) obtained from the Sentinel-2A satellite. The green, red, and black colors represent correctly detected positive samples, incorrectly detected positive samples, and background areas, respectively. (Best to view in color.)

$$\text{mDice} = \frac{1}{N}\sum_{i=1}^{N}\frac{2\times \text{TP}_i}{2\times \text{TP}_i + \text{FP}_i + \text{FN}_i}$$
$$\text{mAcc} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \tag{8}$$

where N represents the number of total classes, which in this case is 2, representing the background and red tide classes. Specifically, TP_i and FP_i denote the true positive and false positive for class i, respectively, representing the number of pixels correctly or incorrectly identified as belonging to class i. FN_i refers to false negative, which is the number of pixels of class i incorrectly identified as another class. TN_i denotes true negative, which is the number of pixels correctly identified as not belonging to class i.

C. Comparison With the SOTA Methods

In the comparative experiments with SOTA methods, we selected several commonly used methods in semantic segmentation tasks, including U-Net [19], Deeplabv3 [67], Deeplabv3+ [68], and Swin-UNet [35]. Additionally, we included methods specifically designed for red tide segmentation tasks, such as GF1_RI [24] and RDU-Net [25]. Furthermore, we incorporated a recent method based on selective state-spaces models (SSMs) [69] for comparative testing. For the binary quantization comparison, we also evaluated several typical methods [53], [54], [55] to assess performance. To validate the robustness of our method across multiple scenarios and datasets, we conducted experiments using remote sensing images collected from different satellite platforms. The differences in acquisition time and location for Areas A and B are detailed in Table II, with HAB conditions in Area B significantly differing from those in Area A. A detailed comparison of our method with SOTA methods in Areas A and B is presented in Tables III and IV, respectively. Additionally, a visual comparison of experimental results is shown in Figs. 5 and 6. These results demonstrate that our method achieves excellent segmentation performance across diverse image scenarios, highlighting its effectiveness on various satellite platforms and conditions. This robust performance underscores the adaptability and accuracy of our approach in different remote sensing environments.

In the comparison experiment at Test Area A, conventional CNN methods [19], [67], [68] serve as baselines. While these methods are effective in general segmentation tasks, they struggle to capture global context, which is crucial for accurate red tide segmentation in complex remote sensing images. The GF1_RI method, which utilizes radiometric indices, performs poorly, achieving an mIoU of 47.69% and an mDice of 49.68%. This highlights that relying solely on the traditional fixed index-based method set according to the spectrum is insufficient for achieving the fine-grained segmentation required for red tide monitoring in large-scale and multiscenario environments. By employing a dynamically shifting binary quantization ViT block, our method achieves the highest performance, with an mIoU of 68.41%, an mDice of 78.32%, and an mAcc of 87.34%. The proposed binary ViT framework outperforms methods utilizing the same framework. Furthermore, compared to models based on the standard ViT framework, the quantized version significantly reduces parameter usage, thereby enhancing both inference and deployment efficiency. The corresponding quantization loss of the binary quantized model with its counterpart has been further examined through subsequent ablation experiments. The experimental results in Test Area B demonstrate that our method has achieved superior performance compared with similar methods, with mIoU, mDice, and mAcc of 56.39%, 62.24%, and 60.18%, respectively. This indicates that our method exhibits strong adaptability across different red tide monitoring scenarios. Additionally, it can be observed that the performance of the GF1_RI method, based on a specific index design, is superior in Area B compared to Area A. This discrepancy highlights the intrinsic limitations of

TABLE III
COMPARISON OF THE SOTA METHODS FOR THE TEST AREA A

Method	Base	Param(M)	mIoU	mDice	mAcc
GF1_RI [24]	Index	-	47.69	49.68	50.19
U-Net [19]	CNN	34.5	65.05	75.29	88.79
DeeplabV3 [67]		28.9	64.92	75.03	85.42
DeeplabV3+ [68]		41.2	63.95	73.47	73.07
RDU-Net [25]		34.7	63.38	72.56	68.23
VM-UNet [69]	SSM	27.4	66.97	76.94	85.22
Swin-UNet [35]	ViT	27.1	66.99	46.47	72.43
Swin-ViT [32]		58.9	56.43	67.16	89.39
Vanilla Bi-ViT [55]	Bi-ViT	13.4	52.13	62.37	81.12
BiViT [54]		15.4	58.61	69.25	**89.74**
BinaryViT [53]		22.6	65.13	74.93	78.56
Ours-Binary		22.6	**68.41**	**78.32**	87.34

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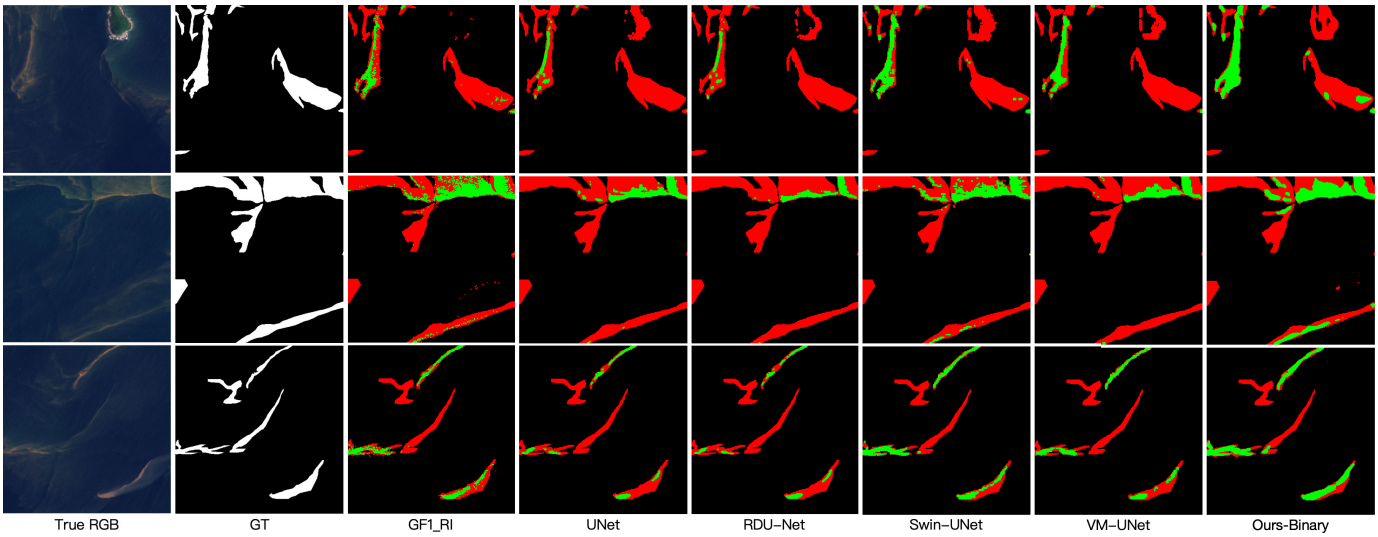


Fig. 6. Visualization of the results comparing different methods applied to the test imagery (referred to as Test Area B) from PlanetScope satellite. The green, red, and black regions indicate correctly detected positive samples, incorrectly detected positive samples, and background areas, respectively. (Best to view in color.)

TABLE IV
COMPARISON OF THE SOTA METHODS ON THE STUDY AREA B

Method	Base	Param(M)	mIoU	mDice	mAcc
GF1_RI [24]	Index	-	54.48	59.01	55.36
U-Net [19]	CNN	34.5	51.74	54.63	53.51
DeeplabV3 [67]		28.9	50.93	52.99	52.01
DeeplabV3+ [68]		41.2	50.74	52.69	51.87
RDU-Net [25]		34.7	51.38	53.83	52.72
VM-UNet [69]	SSM	27.4	52.66	56.14	54.30
Swin-UNet [35]	ViT	27.1	54.49	59.31	57.64
Swin-ViT [32]		58.9	50.82	52.71	51.72
Vanilla Bi-ViT [55]	Bi-ViT	13.4	50.03	51.26	50.87
BiViT [54]		15.4	49.61	50.44	50.38
BinaryViT [53]		22.6	49.29	49.65	50.00
Ours-Binary		22.6	**56.39**	**62.24**	**60.18**

index-based approaches in diverse regional contexts. Relying heavily on predefined thresholds and fixed band combinations, index-based methods need to be frequently calibrated under specific environmental conditions. These static parameters will inevitably limit the adaptability of such methods, hindering their robust performance in complex environmental regions or under changing environmental conditions. To further elucidate this phenomenon, additional ablation experiments are detailed in Section IV-D.

The experimental results demonstrate that our proposed method for red tide segmentation achieves significant improvements over the existing approaches. By leveraging the ViT mechanism, the method effectively addresses the limitations of conventional CNN-based methods in capturing global context, enabling superior global feature aggregation. The integration of multiple spectral bands through a bimodal multispectral combination further enhances feature extraction, resulting in finer segmentation granularity and improved accuracy. Incorporating the UPP module ensures robust feature extraction, refining the segmentation process and boosting overall performance. Additionally, our dynamic offset binary quantization approach reduces parameter redundancy and enhances computational efficiency without compromising segmentation quality. This combination of advanced techniques results in a robust and accurate solution for remote sensing image segmentation, particularly in the context of red tide segmentation.

D. Ablation Study

First, we designed a set of comparative experiments using combinations of visible and other multispectral bands as inputs to validate the effectiveness of the proposed dual-modal multispectral fusion approach in enhancing semantic segmentation performance. Specifically, we considered some widely used indices in remote sensing, including the NDVI and the normalized difference water index (NDWI). The calculation method for the NVDI index is presented in (1) and the NDWI index is presented in the following equation:

$$\mathrm{NDWI} = (B_{\mathrm{Green}} - B_{\mathrm{NIR}})/(B_{\mathrm{Green}} + B_{\mathrm{NIR}}). \tag{9}$$

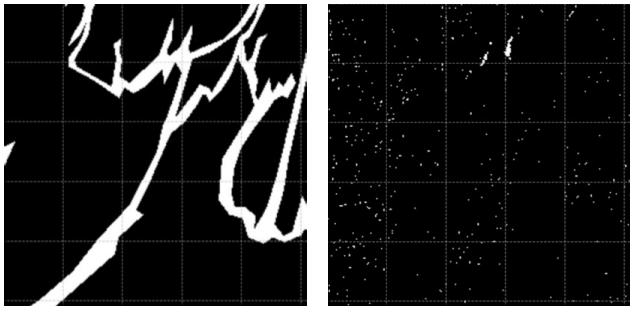
In the comparative experiments, we compared the results of using only True RGB inputs against that of combining True RGB with different spectral band indices. Specifically, this involved comparing the results obtained from RGB images alone with those achieved by concatenating spectral band indices with RGB inputs. The detailed experimental results, shown in Table V, reveal that the multispectral combination inputs outperform the RGB-only inputs. This outcome highlights the enhanced feature representation capability provided by bimodal multispectral data. While visible light images offer rich color and texture information, spectral band data contribute physical characteristics beyond the visible spectrum, enabling the model to perform effectively in more complex environments. Moreover, the bimodal data input significantly enhances the model's generalization ability. With inputs constrained to a single data source, the model may exhibit heightened sensitivity to specific types of interference or noise. However, combining diverse data sources allows the model to learn across a broader range of scenarios, thereby

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TABLE V

COMPARISON OF USING TRUE RGB AND AN EXPONENTIAL INDEX COMPOSED OF DIFFERENT SPECTRAL BANDS AS PAIRWISE COMBINATION INPUTS TO THE MODEL ON THE RESULTS OF RED TIDE SEGMENTATION

Input Modal	mIoU	mDice	mAcc
True RGB	56.43	67.16	89.39
+ NDVI	**60.71**	**71.12**	**90.90**
+ NDWI	60.45	70.06	75.37



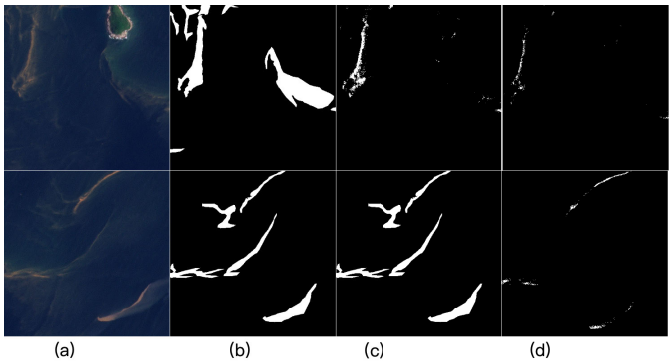
(a) (b)

Fig. 7. Visual comparison between the prediction masks generated using the GF1_RI index and the corresponding GT masks. (a) GT. (b) Prediction.

enhancing robustness in various imagery applications collected from diverse satellite platforms.

Furthermore, despite being specifically designed for red tide segmentation, the GF1_RI method [24] exhibited significant performance degradation in our experiments, as clearly seen in the visual comparison between the predictions from the GF1_RI index and the GT mask shown in Fig. 7. While index-based approaches are computationally efficient and demonstrate high sensitivity in certain conditions, their fixed-threshold mechanism, akin to hard-thresholding, lacks the necessary adaptability to varying environmental conditions and imagery from different satellite platforms. This limitation is particularly pronounced in Test Area A, where such a rigid approach struggles to maintain accuracy across diverse scenarios.

Additionally, given the relatively strong performance observed in Test Area B, we complemented the results of the ablation test by visualizing the impact of threshold setting variations on the segmentation results. This was tested by mapping the index-based radiance values used in the GF1_RI method at different threshold settings to the binarized segmentation mask, thereby illustrating their effect on performance, as illustrated in Fig. 8. It is evident that the settings of different hard threshold values influence the results. However, manually adjusting this value for each detection region is impractical. This limitation aligns with the inherent constraints of the index-based method, which assesses the performance based on single-scene imagery. Consequently, the index-based method is only suitable for specific regions and categories of monitoring tasks, with limited generalizability across datasets from different satellite platforms. While index-based methods [24] undoubtedly have their limitations, the multisource data they utilize are valuable for feature representation. Therefore, we incorporate the spectral information used by index-based methods as an input source into multispectral feature sets, providing complementary information for improved red tide segmentation.



(a) (b) (c) (d)

Fig. 8. Visualization results of the impact of mapping the index-based radiance values at different threshold settings to the segmentation mask. (a) Input. (b) GT. (c) Thresh = 0.3. (d) Thresh = 0.5.

TABLE VI

PERFORMANCE COMPARISON OF DIFFERENT PARAMETER CONFIGURATIONS ON Swin-UNet [35] WAS CONDUCTED, WITH THE PATCH WINDOW SIZE CONSISTENTLY SET TO 8 ACROSS ALL EXPERIMENTS. THE "W/O" (WITHOUT) AND "W" (WITH) DENOTE THE ABSENCE OR PRESENCE OF THE UPP MODULE DESIGN, RESPECTIVELY

Module	Crop Size	Max Epoch	mDice	mAcc
w/o	256	150	73.49	88.68
w/o	256	200	67.79	84.48
w/o	512	150	46.67	72.43
w/o	512	200	38.84	72.09
w	256	150	**79.06**	89.34
w	512	150	75.13	**92.34**

Building on the results of our previous experiments, we aimed to enhance the feature extraction capabilities of the backbone component in the Swin-ViT structure. Initially, it was observed that employing the Swin-UNet [35] architecture alone led to severe serration in the segmentation results due to the inherent mechanism of the ViT, as illustrated in Fig. 9. This serration posed significant challenges in accurately delineating the intricate edge variations characteristic of red tide phenomena in aquatic environments. To mitigate this issue, we designed a series of comparative experiments aimed at evaluating the influence of various adjustable parameters, such as crop size and epoch size.

The hyperparameters were kept constant in these experiments, with the patch window size fixed to 8. The experimental results, presented in Table VI, reveal that crop size has a negligible effect on the segmentation results, while increasing the epoch size exacerbates the overfitting issue. Moreover, the performance in the table reflects the overfitting phenomenon, where the model's performance on the test set significantly decreases as the number of training epochs increases. Based on these observations, we have limited the maximum number of training epochs to effectively mitigate the overfitting and improve the model performance. We have also conducted a series of ablation experiments on the Max Epochs setting, with the specific results shown in Fig. 10. It can be observed that model performance gradually improves and reaches an optimal

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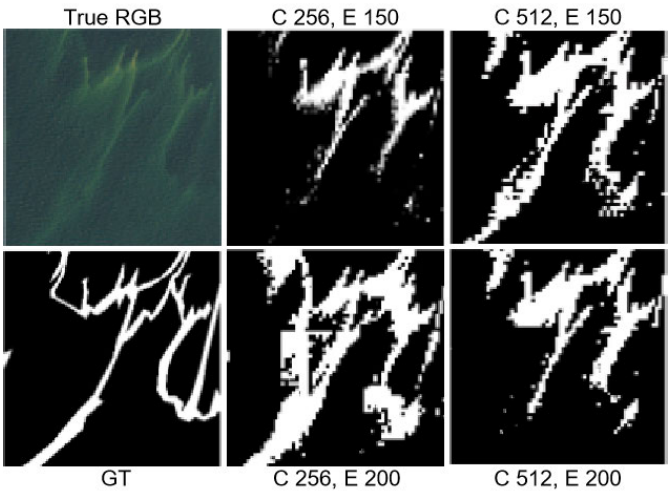


Fig. 9. Visualization results of the serration phenomenon observed in the results of Swin-UNet with different parameter designs, where the UPP module is not included. "C" denotes the crop size, and "E" refers to the training epoch.

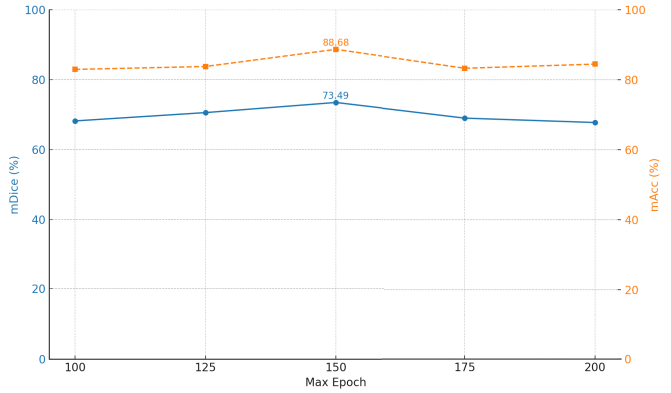


Fig. 10. Impact of different Max Epoch settings during the training phase on model performance.

state between 100 and 150 epochs. Beyond 150 epochs, the model performance begins to decline due possibly to the increasing issue of overfitting. Therefore, we set the maximum number of training epochs to 150 in our experiments and used this as a benchmark for further comparisons of the crop size.

For the quantization aspect, we considered the magnitude of the influence of binary network modules with different design structures on the Swin Transformer-based segmentation framework. This investigation aimed to understand how these different structures influence the performance gap when compared to the FP32 structure and the design of the binary structure with the minimum gap loss is excavated, enabling the possibility of binary inference while maintaining the segmentation accuracy of the model.

The vanilla binary ViT structure, illustrated in Fig. 11(a), represents the simplest approach to binarizing the model. This process involves a direct replacement of the MHA and feed-forward network (FFN) components in the Transformer architecture with their binary equivalents. As evident from the experimental results, this structure leads to significant performance degradation. The main issue with this design is its inability to preserve the precision required for effective feature

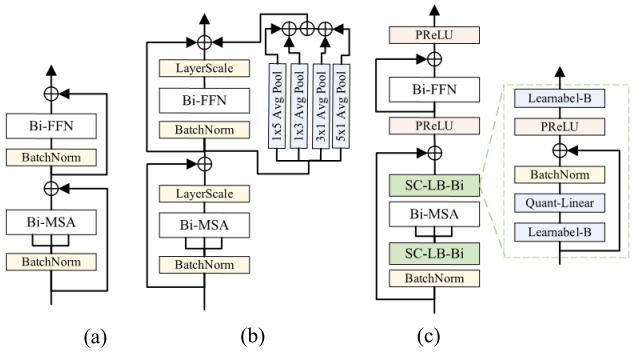


Fig. 11. Comparison of ours and conventional binary ViT architecture. (a) Vanilla Bi-ViT block. (b) RS-Pool Bi-ViT block. (c) Ours Bi-ViT block.

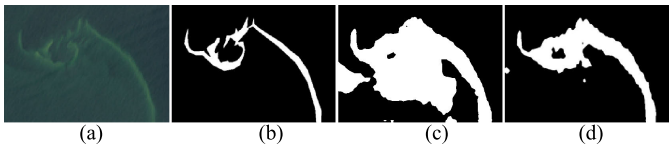


Fig. 12. Impact of the multipooling structure on local boundary details in high-precision segmentation. (a) Input. (b) GT. (c) Multilayer pooling. (d) Ours—binary.

extraction, which is essential for high-sensitivity, fine-grained visual tasks. Consequently, the network struggles to learn and model subtle differences in the data, resulting in suboptimal performance, particularly in applications that require capturing complex patterns and intricate details.

As illustrated in Fig. 11(b), the multiscale aware multipooling structure within the binary ViT block enhances the model's ability to perceive images at different scales and details by performing pooling operations at various scales. This helps capture diverse features, improving the model's recognition accuracy in complex scenes. However, the multipooling structure can lead to excessive smoothing of features. As noted in previous studies [70], [71], downsampling-based pooling operations are inherently lossy. The primary purpose of the pooling layer is to reduce the spatial dimensions of the feature map, thereby improving computational efficiency and facilitating the extraction of higher-level semantic features. However, the pooling process may reduce the spatial resolution of the feature map by aggregating pixel values within local receptive fields (e.g., the maxima or averages). Consequently, this operation can inevitably result in the gradual loss of local detail information, which will in turn affect the segmentation accuracy.

When examining the segmentation results in detail, it becomes evident that the incorporation of a multilayer pooling structure tends to blur the edges, as illustrated in Fig. 12. This smoothing effect, caused by the multiple pooling layers, results in inaccurate segmentation in certain detail-rich regions. During the downsampling process, critical local details in these areas are smoothed out, diminishing the model's ability to accurately delineate the boundaries and causing a decline in the segmentation accuracy. Consequently, while the multipooling structure can enhance feature recognition across different scales, it may also lead to the loss

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TABLE VII
COMPARISON OF DIFFERENCE BINARY QUANTIZATION BLOCKS FOR THE TEST AREA A

Method	#bit (W/A)	mIoU	mDice	mAcc
Ours-FP32	32/32	69.35	79.22	89.76
Vanilla	1/1	52.13	62.37	81.12
Multi-Pooling		65.13	74.93	78.56
Ours		**68.41**	**78.32**	**87.34**

TABLE VIII
QUANTITATIVE EXPERIMENTAL RESULTS ON THE EXECUTION TIME OF OUR MODEL WITH THE COMPARISON METHODS FOR THE TEST AREA A

Method	Backbone	Execution Time (per iteration)	Δ (Improvements)
Swin-UNet [35]	ViT-based	0.0863 s	-
Swin-ViT [32]		0.0545 s	-36.9%
Vanilla Bi-ViT [55]	Binary ViT-based	0.0290 s	-66.4%
BinaryViT [53]		0.0636 s	-26.3%
Ours-Binary		0.0393 s	-54.5%

of critical local boundary information. This limitation is particularly significant in tasks that require precise boundary delineation and the preservation of intricate structural details, such as the segmentation of fine-grained objects or applications demanding high spatial resolution.

Our DoBi-ViT block design is an enhanced version based on the W-MSA and SW-MSA mechanisms of the Swin Transformer, as shown in Fig. 11(c). By incorporating the SC-LB-Bi module before and after the W-MSA, the features entering the Bi-W-MSA can achieve better binarization processing results, mitigating performance loss during the binarization phase. Additionally, our model is tailored to specific tasks and does not use a complete binarization approach that would degrade model performance. Instead, it reduces the redundant parameter bandwidth of common modules at key points, enabling potential embedded deployment of the model.

Based on the analysis of the aforementioned structural variations, we conducted a series of ablation experiments to assess the performance of the proposed various binarized ViT blocks. The detailed experimental results are presented in Table VII. All quantization schemes were implemented using our FP32 model (Ours-FP32), with the vanilla binary file serving as the baseline for performance comparison. Additionally, the multipooling block method, which enhances feature extraction and segmentation accuracy by aggregating information from various image regions, is included as a comparative case. Our proposed method outperforms both the vanilla and multipooling methods, achieving mIoU, mDice, and mAcc scores of 68.41%, 78.32%, and 87.34%, respectively. Notably, compared to the multipooling method, our approach demonstrates a 3.38% increase in mIoU, a 3.39% increase in mDice, and an 8.18% increase in mAcc. This superior performance is attributed to integrating dynamic magnitude offset binary quantization in our method, which effectively reduces parameter redundancy and enhances computational efficiency while maintaining high segmentation quality.

To comprehensively evaluate the performance of our model during the inference phase, we compared it with Swin-UNet [35], Swin-ViT [32], Vanilla Bi-ViT [55], and BinaryViT [53], using the ViT-based Swin-UNet as the baseline. This comparison effectively highlights the differences in inference efficiency among various Transformer-based models, particularly when applied to large-scale, high-resolution imagery. As shown in Table VIII, the inference time for Swin-UNet, serving as the baseline, was 0.0863 s. With an execution time of 0.0545 s per iteration, Swin-ViT demonstrated a notable 36.9% improvement in the inference efficiency over the Swin-UNet. In addition, Binary ViT-based models, including the Vanilla Bi-ViT and Binary ViT, showed different levels of inference efficiency. Vanilla Bi-ViT achieved an execution time of 0.0290 s, representing a 66.4% improvement over Swin-UNet, while BinaryViT exhibited an inference time of 0.0636 s, resulting in only a 26.3% improvement. The observed variation in Binary ViT's performance may be attributed to the multilayer average pooling operations it used, resulting in extra computational bottlenecks and impact the inference speed.

By incorporating further optimizations to the Binary ViT architecture, our DoBi-SWiP-ViT achieved the competitive inference performance with an execution time of 0.0393 s per iteration, a 54.5% improvement over the Swin-UNet. This significant performance improvement underscores that our model can not only offer significant advantages in the inference speed but also effectively reduce the inference latency while maintaining high segmentation accuracy. The comparative analysis has clearly validated the superiority of our model in terms of inference efficiency, strong applicability, and great potential for real-world applications.

V. CONCLUSION

This article proposes a binary quantization ViT for the effective segmentation of red tide in multispectral remote sensing imagery, addressing the challenges of monitoring red tide hazards. By integrating bimodal and cross-level feature fusion UPP modules along with an efficient binarization mechanism for ViT, our DoBi-SWiP-ViT facilitates the effective segmentation of red tides in remote sensing imagery. This approach not only seamlessly integrates the global and local semantic information to ensure the extraction of fine-grained semantic features but also offers a computationally efficient inference solution for transformer frameworks, which are typically characterized by high parameter consumption. Furthermore, we have curated a dataset for segmenting HABs in seawater bodies, comprising high-resolution imagery obtained from sensors on open-access satellite platforms. Based on this dataset, we conducted comparative analyses with SOTA methods and various recently proposed techniques for red tide segmentation. The results demonstrate the superior segmentation performance of our proposed DoBi-SWiP-ViT, achieving finer segmentation granularity compared to SOTA methods. The integration of bimodal and cross-level feature fusion modules within the ViT framework effectively balances global and local semantic information, which is essential for accurately segmenting complex and varied patterns in remote sensing imagery. Moreover, the introduction of a dynamic magnitude offset binary quantization mechanism effectively reduces the computational burden of the ViT, offering a

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lightweight solution without sacrificing accuracy. This is particularly important in large-scale remote sensing applications, where computational resources are often limited. With the reduced revisiting cycles of remote sensing satellites, this study aims to enable early monitoring and prompt response to red tide outbreaks, thereby enhancing the speed of protection and mitigation of HABs.

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